



SYDNEY BOYS HIGH
MOORE PARK, SURRY HILLS

2003
TRIAL
HIGHER SCHOOL CERTIFICATE

Mathematics

General Instructions

- Reading time — 5 minutes
- Working time — 3 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided on the back of this
- All necessary working should be shown in every question

Total marks — 120

- Attempt questions 1–10
- All questions are of equal value, the mark value is shown beside each part.
- Hand up your paper in three parts:
Section A, Questions 1, 2, 3, & 4;
Section B, Questions 5, 6, and 7;
Section C, Questions 8, 9, and 10.

Examiner: P.Bigelow

Note: This is an assessment task only and does not necessarily reflect the content or format of the Higher School Certificate.

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln(x + \sqrt{x^2 - a^2}), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln(x + \sqrt{x^2 + a^2})$$

NOTE : $\ln x = \log_e x, \quad x > 0$

Total marks – 120

Attempt Questions 1–10

All questions are of equal value

Answer each Section in a SEPARATE writing booklet. Extra writing booklets are available.

Section A

Marks

Question 1 (12 marks) Use a SEPARATE writing booklet.

- (a) Evaluate, correct to three significant figures: 2

$$\frac{4.73 + 3 \cdot 1^2}{5.6 \times 9.4}$$

- (b) Solve $x^2 = 10x$ 2

- (c) Differentiate:

(i) $4 - 3x^2$ 2

(ii) xe^x 2

(iii) $\frac{\sin x}{x}$ 2

- (d) Write down a quadratic equation with roots $3 + \sqrt{2}$ and $3 - \sqrt{2}$. 2

Section A continued

Marks

Question 2 (12 marks)

(a) Simplify $\frac{x^2 - 4x}{x - 4}$. **2**

(b) Convert $\frac{3\pi}{5}$ to degrees. **1**

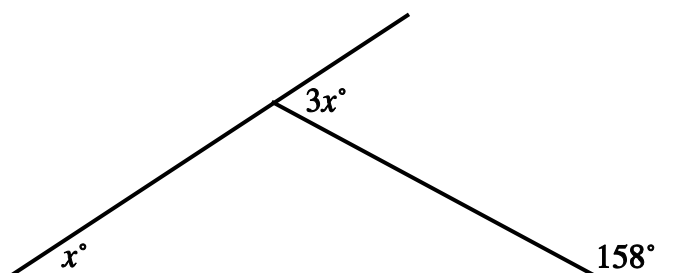
(c) If $\sqrt{75} + \sqrt{80} - \sqrt{12} = 4\sqrt{c} + a\sqrt{3}$, find a and c . **2**

(d) Find (i) $\int \frac{dx}{1+x}$ **1**

(ii) $\int_0^1 \frac{4}{e^{2x}} dx$ **2**

(e) Find the equation of the normal to $y = (3x+4)^3$ at the point where $x = -1$. **2**

(f) **2**



In the diagram above, find the value of x .

Section A continued**Marks****Question 3** (12 marks)

- (a) Given the function $f(x) = \sqrt{64 - x^2}$

state the (i) domain

1

and (ii) range of the function.

1

- (b) Solve this pair of equations simultaneously.

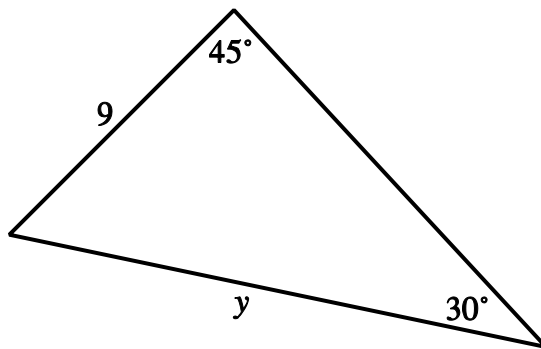
2

$$x + 3y = -7$$

$$4x - y = -2$$

- (c)

Find the exact value of y .

2

- (d) If $\int_0^a (x-3)dx = -4$, find the value(s) of a .

3

- (e) Factorise (i) $16 - a^2$

1

- (ii) $4c^2 + 15c - 4$.

2

Section A continued

Marks

Question 4 (12 marks)

(a) Solve for x :

$$3^x - 3^{x-1} = 54.$$

2

(b) Simplify $\frac{\cos(90^\circ - \theta)}{\sin(180^\circ + \theta)}$.

2

(c) Evaluate.

$$3 + 5 + 7 + 9 + \dots + 81$$

2

(d) If α and β are the roots of $x^2 - 4x + 2 = 0$, find the value of :

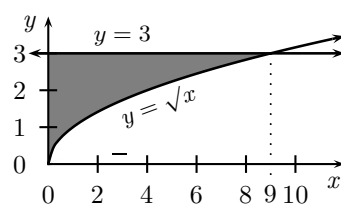
(i) $\frac{1}{\alpha} + \frac{1}{\beta}$

1

(ii) $\alpha^2 + \beta^2$.

2

(e)



3

The diagram shows the area bounded by the y-axis, the curve $y = \sqrt{x}$, and the line $y = 3$. Find the area of the shaded region.

Section B Use a SEPARATE writing booklet.

Marks

Question 5 (12 marks)

(a) Given the parabola $(x+2)^2 = 8(y-1)$, Write down

(i) the co-ordinates of the focus, **1**

(ii) the equation of the directrix. **1**

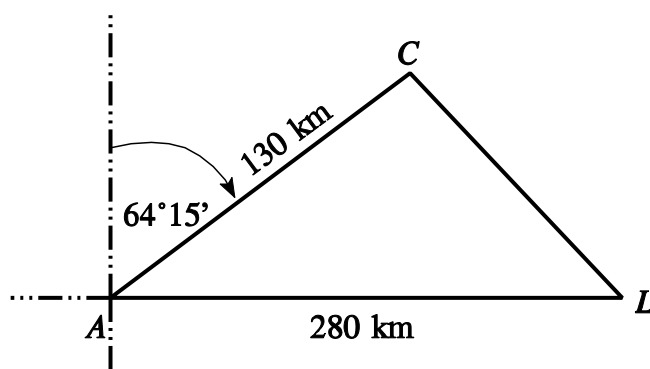
(b) (i) Draw a number plane and mark on it the points $A(4, 3)$, $B(12, -3)$, and $C(10, 7)$. **1**

(ii) Find the equation of the line AB . **1**

(iii) Find the distance of C from the line AB . **2**

(iv) Find the area of the triangle ABC . **2**

(c)



A ship A is 280 km west of a lighthouse L . It travels a distance of 130 km on a bearing of $N64^\circ 15'E$ to a position C .

(i) Calculate the distance from the lighthouse to the ship's position at C . **2**

(ii) Find the bearing of C from the lighthouse L . **2**

Section B continued

Marks

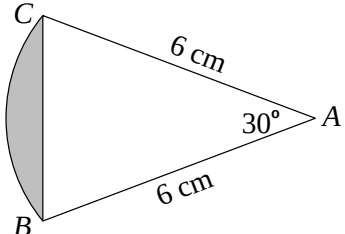
Question 6 (12 marks)

- (a) (i) On the same set of axes, carefully sketch the graphs of $y = \cos x$ and $y = \sqrt{3} \sin x$ where $0 \leq x \leq 2\pi$. **3**
- (ii) Find the x -values of the two points of intersection. **2**
- (iii) Hence solve $\cos x < \sqrt{3} \sin x$ for $0 \leq x \leq 2\pi$. **1**

- (b) The table below shows the values of $f(x)$ for $0 \leq t \leq 2$. **3**

t	0	0.5	1	1.5	2
$f(t)$	0	0.32	0.39	0.35	0.26

Use the Trapezoidal Rule with 5 function values to approximate $\int_0^2 f(t) dt$ correct to 1 decimal place.

- (c)  In the diagram $\angle CAB = 30^\circ$.
 CB is a circular arc of radius 6 cm.

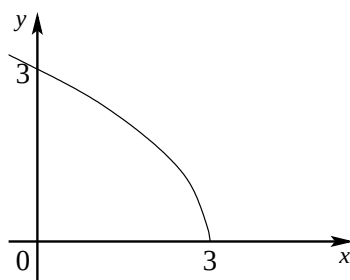
- (i) Find the area of $\triangle ABC$. **1**
- (ii) Calculate the exact area of the shaded region. **2**

Section B continued

Marks

Question 7 (12 marks)

- (a) Consider the curve $y = 2x^3 + 3x^2 - 12x - 9$.
- (i) Find all stationary points and determine their nature. 2
- (ii) Find any points of inflexion. 2
- (iii) Sketch the curve for $-3 \leq x \leq 3$, showing the y-intercept. 2
- (iv) For what values of x is the curve increasing and concave down? 2
- (b) A solid is formed by rotating the part of the curve $y = \sqrt{9-3x}$ between the points $(3, 0)$ and $(0, 3)$ about the y-axis, as shown in the diagram below. Find the volume of the solid. 2



- (c) The volume $V \text{ cm}^3$ of a balloon is increasing such that its volume at any time t seconds is given by $V = \frac{\pi t^3}{3} - \frac{\pi t^2}{6} + \frac{1}{2}$. Find the rate at which the volume is increasing when $t = 3$. 2

Question 8 (12 marks)

- (a) A football club held a raffle to raise money for the end-of-season trip, 100 tickets were sold and two prizes were offered. Two tickets were drawn without replacement to determine the prize-winners.

Frank bought some of the tickets. The probability that he won both prizes was $\frac{2}{275}$. Find:

(i) The number of tickets bought by Frank. **2**

(ii) The probability of his winning at least one prize. **2**

- (b) An Electrical Goods store has a special deal on digital wide-screen TVs. It is offering a loan of \$12 000 with an interest free period of 12 months. From then on, interest is charged at the rate of 12% p.a. monthly reducible.

Patrick takes out the loan and agrees to repay it over four years by making 48 equal monthly repayments of \$ M .

Let \$ A_n be the amount owing after n repayments.

(i) Find an expression for A_{12} . **1**

(ii) Show that $A_{14} = (12\,000 - 12M) \times 1.01^2 - M(1 + 1.01)$. **2**

(iii) Find an expression for A_{48} . **2**

(iv) Find the value of M . **3**

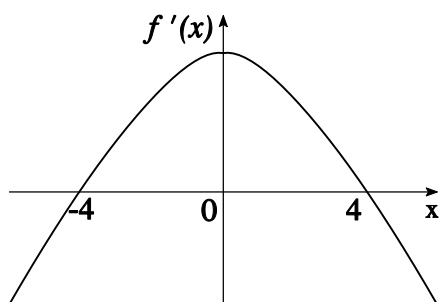
Section C continued

Marks

Question 9 (12 marks)

(a) Solve $\log_3 x - \log_3(x-2) = \frac{2}{3}\log_3 27$. 3

(b) The diagram shows the graph of the gradient function for the curve $y=f(x)$.



(i) What type of point occurs on $y=f(x)$ at $x=4$? Justify your answer. 2

(ii) If $f(4) = 6$ and $f(-4) > 0$, sketch $y=f(x)$. 2

(c) A particle moves with an acceleration given by $f = \sqrt{t} - \frac{1}{\sqrt{t}}$. Initially the particle is moving at $\frac{4}{3} \text{ m.s}^{-1}$ and is $\frac{4}{3} \text{ m}$ to the right of O .

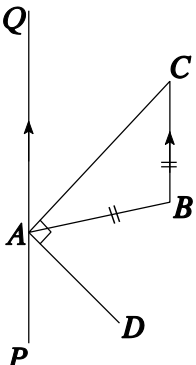
(i) Express the velocity v in terms of t . 2

(ii) Find the displacement x when $t=1$. 3

Section C continued

Marks

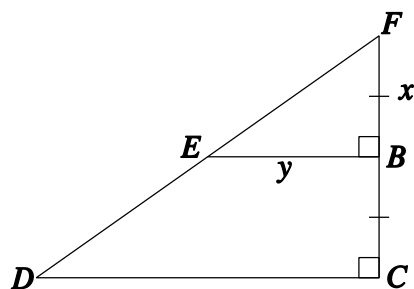
Question 10 (12 marks)

(a)  In the diagram ABC is a triangle with $AB = BC$. The line PQ passes through A parallel to BC , and the line AD is perpendicular to AC .

(i) Prove that AC bisects $\angle QAB$. 2

(ii) Deduce that AD bisects $\angle PAB$. 2

(b)



Farmer George wishes to establish two separate paddocks and sets up his field FCD so that there are fences at FC , DC , and EB as shown on the diagram. The side FD is an existing fence, so no fencing will be required for that side. B is the middle of FC . FB is x metres and EB is y metres.

(i) Write down expressions in terms of x and y for:

(α) BC and DC . 2

(β) The area A of the field FCD . 1

(γ) The amount of new fencing that the farmer would need. 1

- (ii) If the area of the field is 1200 m^2 , show that the length of fencing required is given by: **2**

$$L = 2x + \frac{1800}{x} \text{ metres.}$$

- (iii) Hence find the values of x and y so that the farmer uses the minimum amount of fencing. **2**

END OF THE PAPER

Section A 2 unit

2 1. (a) 0.272

2 (b) $x^2 - 10x = 0$
 $x(x-10) = 0$
 $x = 0 \text{ or } 10$

2 c) i) $-6x$

2 ii) $xe^x + e^x$
 $e^x(x+1)$

2 iii) $\frac{x \cos x - \sin x}{x^2}$

2 d) ~~$(x+3+\sqrt{2})(x-3+\sqrt{2})$~~
 $x^2 - 3x + 8x + 8x^2$
 $x^2 - 6x + 7 = 0$

2 a) $\frac{x(x-4)}{x-4} = x$

1 b) $\frac{3 \times 180}{5} = 108^\circ$

2 c) $5\sqrt{3} + 4\sqrt{5} - 2\sqrt{3}$
 $3\sqrt{3} + 4\sqrt{5} = 4\sqrt{c} + a\sqrt{b}$
 $a=3, c=5$

d) i) $\log(1+x) + c$

ii) $\left[\int 4e^{-2x} \right]_0^1$
 $= \left[-2e^{-2x} \right]_0^1$
 $= \left[-2e^{-2} \right] - \left[-2 \right]$
 $= 2 - \frac{2}{e^2} + 2$

(e) $y = (3x+4)^3$
 $\frac{dy}{dx} = 9(3x+4)^2$

at $x=-1$ $\frac{dy}{dx} = 9$

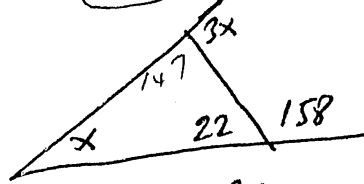
2 $(-1, 1)$ $9y - 9 = 9(x+1)$

$9y - 9 = 9x + 9$

$9x - 9y + 18 = 0$

$x + y - 8 = 0$

(f)



$x + 22 = 3x$

$x = 11$

Q3 ① domain $-8 \leq x \leq 8$

a) ① range $0 \leq y \leq 8$

b) $x + 3y = -7$
 $4x - y = -2$

$13y = -26$

$y = -2$

$x = -1$

$x = -1 \text{ and } y = -2$

c) $\frac{9}{\sin 30} = \frac{4}{\sin 45}$

$9 \times \frac{1}{\sqrt{2}} = \frac{1}{2} \times y$

2 $y = \frac{18}{\sqrt{2}}$

exact $y = 9\sqrt{2}$

d) $\int_0^a (x-3)dx = -4$

77 $\left[\frac{1}{2}x^2 - 3x \right]_0^a = -4$

e) i) $(4-a)(4+a)$

ii) $(4c-1)(c+4)$

Q4 a)

2 $3^x - 3^{x-1} = 54$

$3^x \left[1 - \frac{1}{3} \right] = 54$

$3^x = 81$

$x = 4$

b) $\frac{\sin \theta}{-\sin \theta} = -1$

c) $a=3, d=2$
 $81 = 3 + (n-1) \times 2$

2 $81 = 3 + 2n - 2$
 $n = 40$

d) $\alpha + \beta = 4$
 $\alpha\beta = 2$

1 $\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta} = \frac{4}{2}$

2 ii) $(\alpha + \beta)^2 - 2\alpha\beta$
 $16 - 4 = 12$

e) $A = \int_0^3 x^2 dy$

$A = \int_0^3 y^2 dy$

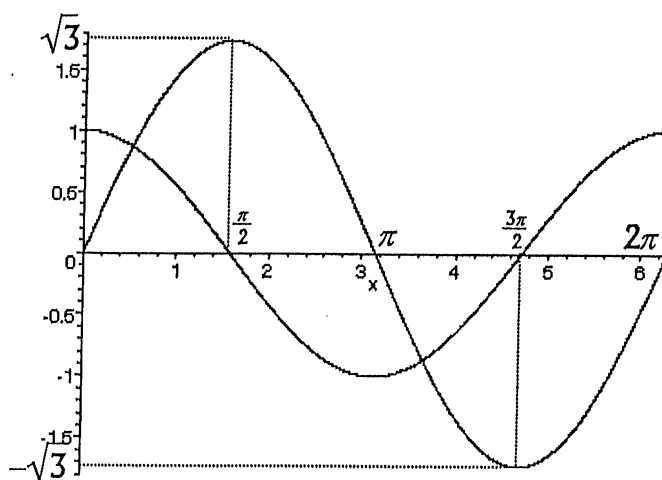
$A = \left[\frac{1}{3}y^3 \right]_0^3$

$A = [9] - [0]$

$A = 9 \text{ units}$

Question 6

(a) (i)



$$(ii) \quad \sqrt{3} \sin x = \cos x \Rightarrow \frac{\sin x}{\cos x} = \frac{1}{\sqrt{3}} \Rightarrow \tan x = \frac{1}{\sqrt{3}}$$

$$x = \frac{\pi}{6}, \frac{7\pi}{6}$$

$$(iii) \quad \frac{\pi}{6} < x < \frac{7\pi}{6} \quad (\text{We need } y = \cos x \text{ to be 'below' } y = \sqrt{3} \sin x)$$

(b)

x	y	w	$y \times w$
0	0	1	0
0.5	0.32	2	0.64
1	0.39	2	0.78
1.5	0.35	2	0.7
2	0.26	1	0.26
			$\Sigma(y \times w) = 2.38$

$$h = 0.5$$

$$\int_0^2 f(t) dt \cong \frac{h}{2} \times 2.38 = 0.6$$

$$(c) (i) \quad \text{Area } \triangle ABC = \frac{1}{2} \times 6^2 \times \sin 30^\circ = 9$$

$$(ii) \quad 30^\circ = \frac{\pi}{6}$$

$$\text{Sector } ABC = \frac{1}{2} \times 6^2 \times \frac{\pi}{6} = 3\pi$$

$$\text{Shaded area} = 3\pi - 9 \text{ cm}^2$$

Question 5

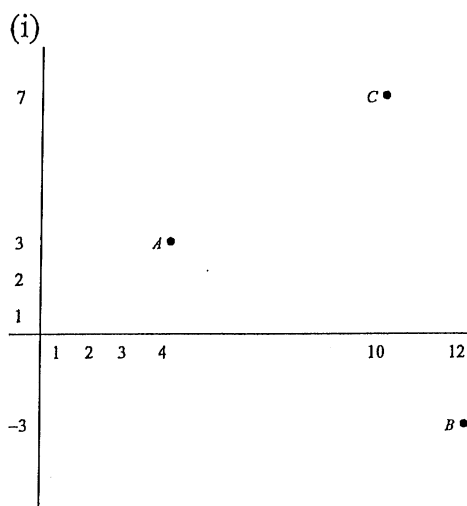
- (a) $(x-p)^2 = 4a(y-q)$ is the parabola with vertex (p, q) and focal length $|a|$

$$(x+2)^2 = 8(y-1) \Rightarrow \text{vertex } (-2, 1) \text{ \& } a = 2$$

(i) focus: $(-2, 1+a) = (-2, 3)$

(ii) directrix: $y = 1 - a = -1$

(b)



(ii) $m_{AB} = \frac{-3-3}{12-4} = -\frac{3}{4}$

$$y - y_1 = m(x - x_1)$$

$$y - 3 = -\frac{3}{4}(x - 4) \Rightarrow y = -\frac{3}{4}x + 3 + 3$$

$$y = -\frac{3}{4}x + 6 \Leftrightarrow 3x + 4y - 24 = 0$$

(iii) $d = \frac{|Ax_c + By_c + C|}{\sqrt{A^2 + B^2}}$

$$3x + 4y - 24 = 0 \Rightarrow A = 3, B = 4, C = -24$$

$$C(10, 7) = (x_c, y_c)$$

$$d = \frac{|3 \times 10 + 4 \times 7 - 24|}{\sqrt{3^2 + 4^2}} = \frac{34}{5}$$

(iv) $AB = \sqrt{(x_A - x_B)^2 + (y_A - y_B)^2} = \sqrt{(4 - 12)^2 + (3 - (-3))^2} = \sqrt{100} = 10$

$$\text{Area} = \frac{1}{2} \times 10 \times \frac{34}{5} = 34$$

(c) (i) $CL^2 = AC^2 + AL^2 - 2 \times AL \times AC \times \cos 25^\circ 45'$

$$CL^2 = 130^2 + 280^2 - 2 \times 130 \times 280 \times \cos 25^\circ 45'$$

$$CL \cong 172.4 \text{ km}$$

(ii) Let $\theta = \angle CLA$, $\angle CAL = 25^\circ 45'$

$$\frac{\sin \angle CLA}{AC} = \frac{\sin \angle CAL}{CL} \Rightarrow \sin \theta = \frac{\sin 25^\circ 45'}{172.4} \times 130$$

$$\therefore \theta = 19^\circ 7'$$

$$\text{Bearing} = 270^\circ + \theta = 289^\circ 7' \text{ T} = N70^\circ 53' \text{ W}$$

$$(b) \quad y = \sqrt{9-3x} \Rightarrow y^2 = 9-3x \Rightarrow 3x = 9-y^2 \Rightarrow 9x^2 = (9-y^2)^2$$

$$\begin{aligned} V &= \pi \int_{y=a}^{y=b} x^2 dy \\ &= \frac{1}{9} \times \pi \int_0^3 9x^2 dy \\ &= \frac{\pi}{9} \int_0^3 (9-y^2)^2 dy \\ &= \frac{\pi}{9} \int_0^3 (81-18y^2+y^4) dy \\ &= \frac{\pi}{9} \left[81y - 6y^3 + \frac{1}{5}y^5 \right]_0^3 \\ &= \frac{\pi}{9} \left(\frac{648}{5} \right) = \frac{72\pi}{5} \text{ c.u.} \end{aligned}$$

$$(c) \quad V = \frac{\pi t^3}{3} - \frac{\pi t^2}{6} + \frac{1}{2} \Rightarrow \frac{dV}{dt} = \pi t^2 - \frac{\pi t}{3}$$

$$t=3, \frac{dV}{dt} = \pi \times 9 - \pi = 8\pi \text{ cm}^3/\text{s}$$

Question 7

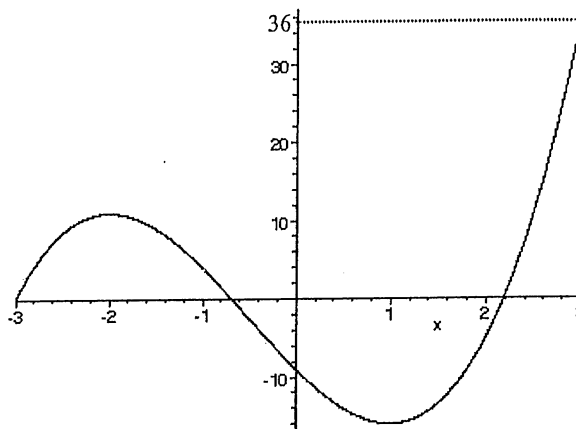
- (a) (i) $y = 2x^3 + 3x^2 - 12x - 9$
 $y' = 6x^2 + 6x - 12 = 6(x^2 + x - 2) = 6(x+2)(x-1)$
 $y'' = 12x + 6 = 6(2x+1)$
 Stationary points when $y' = 0 \Rightarrow x = -2, 1$
 $x = -2 \Rightarrow y = 11, y'' = -18 \Rightarrow (-2, 11)$ is a rel. max.
 $x = 1 \Rightarrow y = -16, y'' = 18 \Rightarrow (1, -16)$ is a rel. min.

- (ii) P.O.I. if $y'' = 0 \Rightarrow x = -\frac{1}{2} \Rightarrow y = -2\frac{1}{2}$ AND a change of concavity

x	-1	$-\frac{1}{2}$	0
y''	-6	0	6

So $(-\frac{1}{2}, -2\frac{1}{2})$ is a P.O.I

- (iii) $x = -3, y = 0$ & $x = 3, y = 36$
 y - intercept $(0, -9)$



- (iv) From the graph, it is increasing and concave down for $x < -2$
 In the domain for (iii) it would be $-3 \leq x < -2$

Question 7 (contd.)

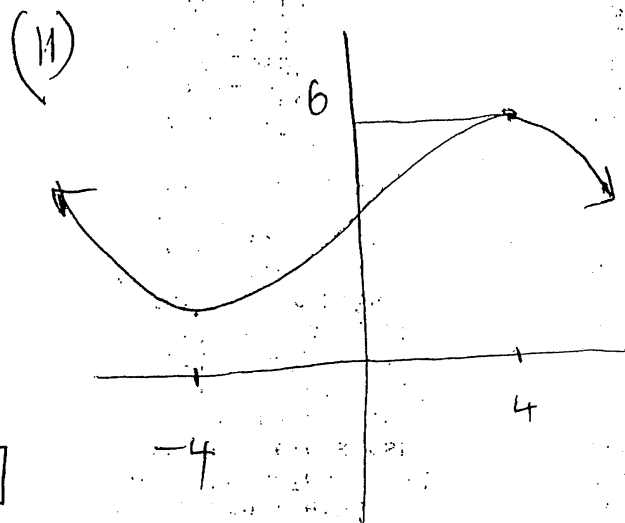
2) (i) Relative Maximum
Turning point.

$$f'(4^-) > 0 \quad +$$

$$f'(4) = 0 \quad -$$

$$f'(4^+) < 0 \quad -$$

[2]



[2]

(c) $f = \sqrt{t} - \frac{1}{\sqrt{t}}$

When $t=0$, $v = \frac{4}{3}$, $x = \frac{4}{3}$

(i) $\ddot{x} = t^{\frac{1}{2}} - t^{-\frac{1}{2}}$

$$\dot{x} = \int (t^{\frac{1}{2}} - t^{-\frac{1}{2}}) dt + C$$

$$= \frac{t^{\frac{3}{2}}}{\frac{3}{2}} - \frac{t^{\frac{1}{2}}}{\frac{1}{2}} + C$$

$$\dot{x} = \frac{2}{3} t^{\frac{3}{2}} - 2\sqrt{t} + C$$

From initial conditions:

$$\frac{4}{3} = 0 + 0 + C$$

[2]

$$\therefore v = \dot{x} = \frac{2}{3} t^{\frac{3}{2}} - 2\sqrt{t} + \frac{4}{3}$$

(ii) $x = \int \left(\frac{2}{3} t^{\frac{3}{2}} - 2t^{\frac{1}{2}} + \frac{4}{3} \right) dt + D$

$$= \frac{2}{3} \frac{t^{\frac{5}{2}}}{\frac{5}{2}} - 2 \frac{t^{\frac{3}{2}}}{\frac{3}{2}} + \frac{4t}{3} + D$$

$$= \frac{4}{15} t^{\frac{5}{2}} - \frac{4}{3} t^{\frac{3}{2}} + \frac{4}{3} t + D$$

By initial conditions $D = \frac{4}{3}$

$$\therefore x = \frac{4}{15} t^{\frac{5}{2}} - \frac{4}{3} t^{\frac{3}{2}} + \frac{4}{3} t + \frac{4}{3}$$

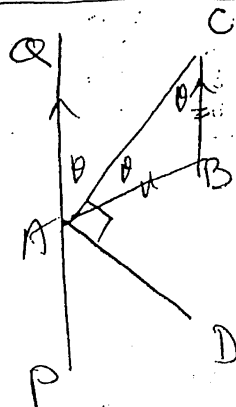
When $t=1$

$$x = \frac{4}{15} - \frac{4}{3} + \frac{4}{3} + \frac{4}{3}$$

$$= \frac{8}{5} \text{ m}$$

[3]

Question 10



(i) Let $\angle ACB = \theta$

$\therefore \angle BAC = \theta$ (Isosceles Δ)

Section C

Question 8

(1) Let n be the no. of tickets

$$p(\text{both}) = \frac{n}{100} \times \frac{n-1}{99} = \frac{2}{275}$$

$$\therefore \frac{n(n-1)}{9900} = \frac{2}{275}$$

$$n^2 - n = 72$$

$$n^2 - n - 72 = 0$$

$$(n-9)(n+8) = 0$$

$$\therefore n = 9 \text{ or } -8$$

(-8 is extraneous)

[2] $\therefore n = 9$

$\therefore$ Frank bought 9 tickets.

(ii) $p(\text{at least 1 prize}) = 1 - p(\text{no prize})$

$$= 1 - \frac{91}{100} \times \frac{90}{99}$$

$$= \frac{19}{110}$$

[2]

b) (i) $A_{12} = 12000 - 12M$

(ii) $A_{13} = A_{12} \times 1.01 - M$

(Monthly interest = 1%)

$$= (12000 - 12M) \times 1.01 - M$$

$$A_{14} = A_{13} \times 1.01 - M$$

$$= (12000 - 12M) \times 1.01^2 - M \times 1.01 - M$$

$$= (12000 - 12M) \times 1.01^2 - M(1 + 1.01)$$

[2]

(iii)

$$A_{48} = (12000 - 12M)(1.01)^{36} - M(1 + 1.01 + \dots + 1.01^{36})$$

AS: $a=1, r=1.01, n=$

$$= (12000 - 12M)1.01^{36} - M \frac{(1.01^{36} - 1)}{1.01 - 1} \quad [$$

(iv)

$$\text{But } A_{48} = 0$$

$$\therefore (12000 - 12M)1.01^{36} = M \frac{(1.01^{36} - 1)}{1.01 - 1}$$

$$12000 \times 1.01^{36} = 12M \times 1.01^{36} + M \frac{(1.01^{36} - 1)}{1.01 - 1}$$

$$12000 \times 1.01^{36} \times 0.01 = 12M \times 1.01^{36} \times 0.01 + M(1.01^{36} - 1)$$

$$= M(12 \times 1.01^{36} \times 0.01 + 1.01^{36} - 1)$$

$$M = \frac{12000 \times 1.01^{36} \times 0.01}{12 \times 1.01^{36} \times 0.01 + 1.01^{36} - 1}$$

$$= \frac{171.692254}{0.60246}$$

$$= \$284.98$$

[3]

Question 9

(a) $\log_3 x - \log_3 (x-2) = \frac{2}{3} \log_3 27$

$$\log_3 \left(\frac{x}{x-2} \right) = \log_3 27^{\frac{2}{3}}$$

$$\frac{x}{x-2} = 9 \quad x \neq 2$$

$$x = 9x - 18$$

$$8x = 18$$

$$x = \frac{18}{8} = \frac{9}{4}$$

[3]

Question 10 (Contd.)

But $\angle QAC = \angle ACB$
 $= \theta$ (alternate $\angle$ s)

$\therefore \angle QAC = \angle CAB$

[2] $\therefore AC$ bisects $\angle QAB$.

(ii) Now $\angle QAP = 180^\circ$ (str. $\angle$)

$\therefore \angle PAD = 180 - 90 - \theta$
 $= 90 - \theta$

But $\angle CAD = \angle CAB + \angle BAD$

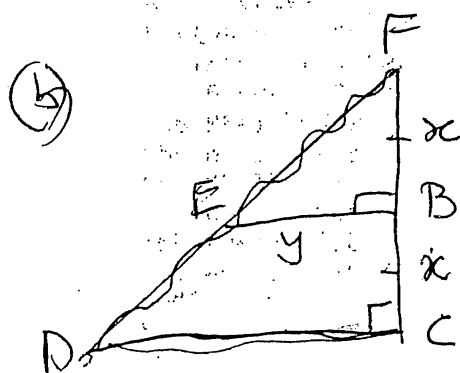
i.e. $90^\circ = \theta + \angle BAD$

$\therefore \angle BAD = 90 - \theta$

$= \angle PAD$

(see above)

[2] $\therefore AD$ bisects $\angle PAB$



(i) $BE = x$

$\frac{DC}{EB} = \frac{FC}{BC}$

$\frac{DC}{y} = \frac{2x}{x}$

(iii) Δ s

$\therefore DC = 2y$ [2]

(B) $A = \frac{1}{2}(2x)(2y)$
 $= 2xy$ [1]

(8) New Fencing

$L = 2x + 3y$ [1]

(ii) $1200 = 2xy$

$\therefore y = \frac{600}{x}$

$\therefore L = 2x + 3 \times \frac{600}{x}$
 $= 2x + \frac{1800}{x}$

(iii) $\frac{dL}{dx} = 2 - \frac{1800}{x^2}$ [2]

$\frac{d^2L}{dx^2} = \frac{3600}{x^3}$

Now $\frac{dL}{dx} = 0$ for $2x^2 - \frac{1800}{x} = 0$

i.e. $2x^2 - 1800 = 0$

$x^2 - 900 = 0$

$(x-30)(x+30) = 0$

$x = 30$ or -30

(-30 is extraneous)

2nd derivative > 0 for $x > 0$.

Minimum L for $x = 30$

$y = 20$